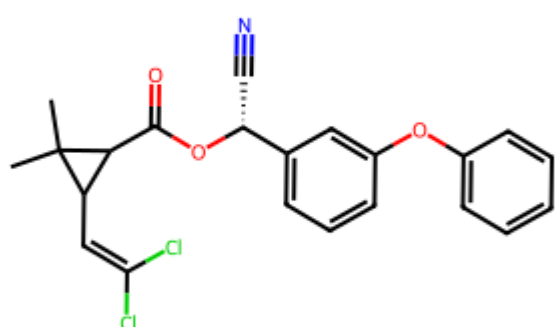
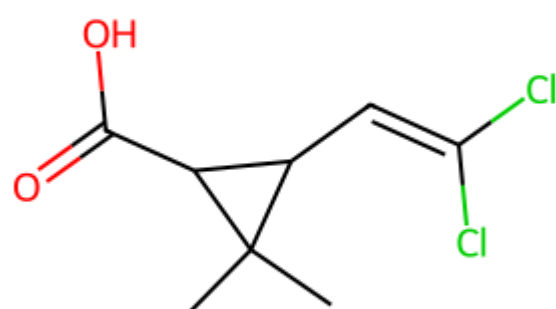




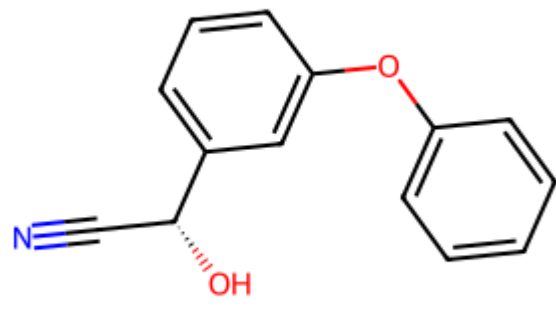
{'analysis_time': 0.0,
'first_solution_time': 1.176588,
'num_calls_rxn_model': 0,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 4.63286949272151,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 4.63286949272151,
'retro_star_value': 4.63286949272151}



Reaction
{'analysis_time': 0.197726,
'first_solution_time': 1.176588,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 4.63286949272151,
'retro_star_reaction_number': 4.63286949272151,
'retro_star_rxn_cost': 0.6655320135269719,
'retro_star_value': 4.63286949272151}



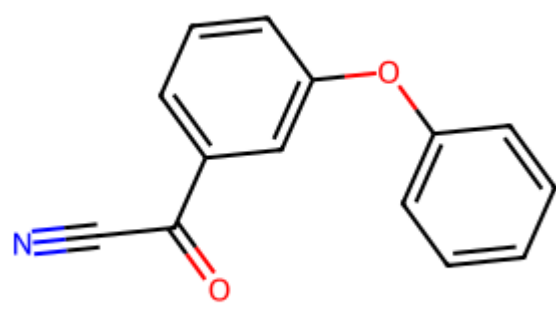
{'analysis_time': 0.197726,
'first_solution_time': 0.197726,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 4.63286949272151}



{'analysis_time': 0.197726,
'first_solution_time': 1.176588,
'num_calls_rxn_model': 1,
'num_calls_value_function': 0,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 3.967337479194538,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 3.967337479194538,
'retro_star_value': 4.63286949272151}



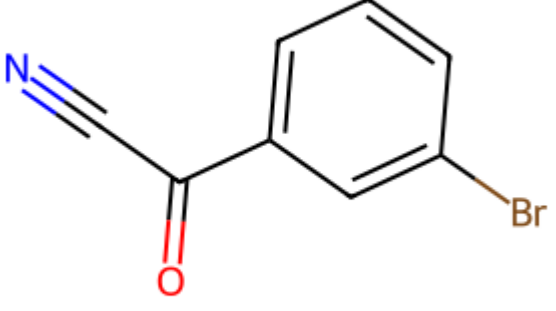
Reaction
{'analysis_time': 0.313483,
'first_solution_time': 1.176588,
'num_calls_rxn_model': 2,
'num_calls_value_function': 42,
'retro_star_min_cost': 3.967337479194538,
'retro_star_reaction_number': 3.967337479194538,
'retro_star_rxn_cost': 0.5430045221302259,
'retro_star_value': 4.63286949272151}



{'analysis_time': 0.313483,
'first_solution_time': 1.176588,
'num_calls_rxn_model': 2,
'num_calls_value_function': 42,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 3.424332957064312,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 3.424332957064312,
'retro_star_value': 4.632869492721509}



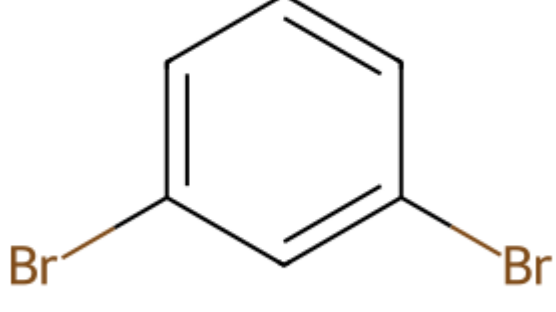
Reaction
{'analysis_time': 0.510215,
'first_solution_time': 1.486844,
'num_calls_rxn_model': 4,
'num_calls_value_function': 110,
'retro_star_min_cost': 3.424332957064312,
'retro_star_reaction_number': 3.424332957064312,
'retro_star_rxn_cost': 2.7646205525906042,
'retro_star_value': 4.632869492721509}



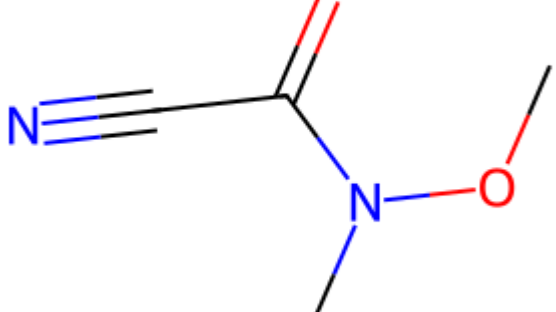
{'analysis_time': 0.510215,
'first_solution_time': 1.486844,
'num_calls_rxn_model': 4,
'num_calls_value_function': 110,
'reaction_number_estimate': 0.0,
'retro_star_min_cost': 0.6597124044737078,
'retro_star_mol_cost': inf,
'retro_star_reaction_number': 0.6597124044737078,
'retro_star_value': 4.63286949272151}



Reaction
{'analysis_time': 1.486844,
'first_solution_time': 1.486844,
'num_calls_rxn_model': 14,
'num_calls_value_function': 515,
'retro_star_min_cost': 2.8302178350764176,
'retro_star_reaction_number': 2.8302178350764176,
'retro_star_rxn_cost': 2.8302178350764176,
'retro_star_value': 6.8033749233242204}



{'analysis_time': 1.486844,
'first_solution_time': 1.486844,
'num_calls_rxn_model': 14,
'num_calls_value_function': 515,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 6.8033749233242204}



{'analysis_time': 1.486844,
'first_solution_time': 1.486844,
'num_calls_rxn_model': 14,
'num_calls_value_function': 515,
'retro_star_min_cost': 0.0,
'retro_star_mol_cost': 0.0,
'retro_star_reaction_number': 0.0,
'retro_star_value': 6.8033749233242204}